

SCI-FLT-MATCHED v0.1

Optimal matched-template map filtering

Owner-assigned package title; not a realization-level optimality claim

Scientific Rationale and Contract

STAGE B DRAFT – ALL AO-001 THROUGH AO-006 OPTIONS UNSELECTED.

No numerical route, implementation, conformity result, response or covariance fidelity, observational validation, achieved performance, readiness, production state, scientific freeze, or Unity status is claimed.

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Part I

Scientific rationale

1 The scientific question

The method answers one narrow question: at each admitted parent-pixel anchor, what is the amplitude of one exact, already-declared template in one exact ordinary-MAP parent? The answer is a map of template amplitude. The method does not decide whether a source exists, choose a candidate, infer a posterior sky, or attach catalog meaning to a peak.

Base v0.1 places one anchor at every exact parent ordinary-MAP pixel center, preserving WCS, frame, grid, shape, indexing, ordering, and the pixel-center convention. It does not interpolate or optimize an anchor. Another output lattice is a separately named successor.

The template is not merely a shape used for smoothing. It is the expected parent-map response per unit amplitude, including its scale, phase, anchor, sampling, support, and calibration/beam lineage. That scaling defines what one unit of the answer means. A template whose unit amplitude denotes point-source flux can support that interpretation only when the exact CAL/BEAM and parent lineages authorize it; another template gives a shape amplitude instead.

2 Why the estimator is normalized

The parent is contracted with the template through a frozen complete coordinate-basis weighting object. Every required pixel or quadrature measure is already inside that bilinear object; there is no second unexplained measure. The same weighting is applied to the template itself, producing the normalization. Dividing these two contractions makes an exact matching template return its supplied amplitude on complete valid support. This unit response is the central scientific identity.

Weighting and uncertainty are deliberately not conflated. If the weighting is the exact authorized inverse covariance on the same population, support, conditioning state, and estimable subspace, the estimator has the familiar inverse-covariance GLS minimum-variance property only after covariance is restricted to local application support and inverted on an exact identifiable estimable subspace. Under a weaker spectral or reliability weight, it can remain matching-template normalized without being minimum variance. In that case the denominator is normalization, not a variance estimate.

3 Why state is explicit and frozen

The transformation must mean the same thing for the signal parent and every compatible NOI realization member. Learn derives candidates and diagnostics from one declared population; Resolve selects the authorized route and freezes its identity; Apply uses it unchanged. Externally declared state enters Resolve. Learning again from each NOI member would change the ensemble method and its uncertainty meaning. New NOI-informed knowledge creates a successor generation rather than rewriting an existing one.

Observation and coadd parents also remain scientifically separate. Filtering an observation and filtering a coadd are two local applications with different parent identities and state generations. This contract neither asserts that filtering commutes with coaddition nor owns a coaddition operation.

4 Why edges are unavailable rather than repaired

Each amplitude depends on complete application support: every parent payload row with a nonzero coefficient in its linear functional. Template-query, state-learning, response-query, and storage supports are recorded separately. If one dependency is absent or invalid, the estimator defined here does not exist at that location. Filling, extending, renormalizing, fitting a background, or learning an edge taper would define another method. Numerical fill is harmless only after erosion proves that no admitted output depends on it.

5 Response and uncertainty are full scientific objects

Unit self-response does not make the response a single symmetric beam or make neighboring sources disappear. For a general linear sky, the expectation is the full response-weighted sum of template amplitudes plus the linear action on any residual mean, offset, gradient, or other deterministic background. The off-diagonal response can be anisotropic, asymmetric, nonstationary, position-dependent, or nonlocal. Reducing it to a kernel, radial profile, or solid angle is allowed only after the required invariances, measure, and valid domain are established. Fixed-state response is not the response of a data-dependent Learn–Resolve procedure. That full-procedure quantity requires an authorized rerun or finite difference and is otherwise unavailable.

Likewise, a conditional covariance product exists only when matching parent covariance authority exists. A field carries the response and covariance of the operator that actually produced it; exact-reference quantities remain comparators. Calibration, learned-state, numerical discrepancy, representation, NOI sampling, and template-mismatch contributions have different populations and conditioning. The normative uncertainty budget keeps them separate rather than producing an unsupported scalar total.

6 Product and governance rationale

The primary science product is one atomic signal bundle: amplitude map plus the facts required to know exactly what it means. Covariance, empirical NOI, response, and state/lineage products are independently atomic companions. Their failure is explicit. Optional failure does not silently alter the signal estimand, while failure of a required signal member prevents publication of a complete signal bundle.

The product also preserves a first-class FLT to FRUIT handoff. It makes every FLT-side scientific fact available or exactly reconstructable without exposing undocumented implementation state. It intentionally says nothing about FRUIT’s future model, iteration, selection, response, uncertainty, or interpretation.

Finally, the contract separates claim layers that are often confused: scientific authority defines the method; conformity concerns a specific implementation; representation fidelity concerns an exact encoding; SCI-VAL evaluates named-use profiles; observational validation and achieved performance are evidence classes; readiness is a governance decision; and production is an operational state. Evidence in one layer does not promote another.

Part II

Shared normative core

The following six modules are imported unchanged into both the scientist- and engineering-facing documents. They are the single normative source for all definitions, equations, assumptions, requirements, predictions, option alternatives, failure rules, and validation consequences.

7 Normative notation and interpretation

7.1 Normative language

“Shall” is a contract requirement. “May” identifies an allowed form whose scientific preconditions still have to be met. “Exact” means equality in the declared mathematical realization. Ordinary finite-precision acceptance is an engineering numerical-conformance question and does not weaken an exact scientific identity. “Unknown” and “unavailable” never mean zero. An option in Section 13 is a decision candidate, not a selected route.

7.2 Collision-free objects, domains, and indexing

The paired readout-coordinate symbols x and r remain reserved to KID authority, and bare Q_p remains reserved to MAP/JINC authority. They are not used for this package’s map-domain anchors or weighting.

A matched-template output anchor is p , a parent-map row or pixel is q , and a separately indexed template anchor is s . The base-v0.1 anchor set $\mathcal{D}_p$ contains exactly one anchor at each parent ordinary-MAP pixel center, with the same WCS, frame, grid, shape, indexing, ordering, and pixel-center convention as the parent apart from exact support restriction. The parent signal sample is m_q . The expected parent response at row q per unit amplitude anchored at p is t_{pq} ; its complete parent-coordinate vector is t_p^{full} . The restriction/extraction matrix for application support is E_p , so

$$m_p = E_p m, \quad t_p = E_p t_p^{\text{full}}.$$

The scientific weighting W_p is the complete coordinate-basis bilinear operator on those restricted vectors. The fixed-state row functional is $L_{p,g}$, the template matrix whose column s is t_s^{full} is T , and $R_{\text{fixed}}(p, s) = L_{p,g} t_s^{\text{full}}$.

All scientific input and output vectors are real. A transform representation may be complex only when it preserves real-field conjugate symmetry and uses the Hermitian adjoint. In the real parent-coordinate basis the adjoint is the ordinary transpose. No stationarity, isotropy, periodicity, flat-sky metric, or Euclidean physical metric follows from the vector notation.

Symbol	Normative meaning
p, q, s	Output anchor, parent row/pixel, and independently indexed template anchor.
m, m_q, m_p	Immutable admitted ordinary-MAP parent vector, one parent sample, and the complete restriction $E_p m$.

Symbol	Normative meaning
t_{pq}, t_p	Parent response per unit amplitude and its complete application-support vector.
E_p	Exact extraction/restriction from parent coordinates to $\mathcal{S}_{\text{apply}}(p)$.
W_p	Frozen, self-adjoint, positive-semidefinite complete bilinear weight; all measure factors required by the scientific contraction are already in it.
n_p	Weighted template-parent contraction $t_p^\top W_p m_p$.
d_p	Weighted template self-contraction $t_p^\top W_p t_p$; exactly real and positive for an admitted anchor.
$\hat{a}_p$	Published fixed-template amplitude n_p/d_p .
$L_{p,g}$	Fixed-state linear functional producing $\hat{a}_p$.
T	Full template-response operator over the declared template-anchor domain.
$R_{\text{fixed}}(p, s)$	Fixed-state response $L_{p,g} t_s^{\text{full}}$.
$R_{\text{FP}}(p, s)$	Full-procedure response when Learn–Resolve is rerun under an authorized perturbation definition; otherwise unavailable.
C_{parent}	Authorized coordinate-basis parent covariance for one exact population and conditioning state.
C_p	Locally restricted covariance $E_p C_{\text{parent}} E_p^\top$.
g	Immutable method/state generation, including requested, effective, resolved, applied, and realized identities.
$\mathcal{V}$	Anchors satisfying every parent, template, application-support, operator, normalization, unit, calibration, response, and selected-route predicate.

7.3 Coordinate-basis measure and covariance convention

This package uses coordinate-basis column vectors and ordinary matrix multiplication. There is no additional measure-weighted bracket in the scientific estimator. Every fixed pixel area, quadrature weight, transform normalization, or metric factor required by a selected realization is part of W_p (and of the corresponding template/covariance coordinate convention).

For a random parent vector with mean $\bar{m}$,

$$C_{\text{parent}} = \mathbb{E}[(m - \bar{m})(m - \bar{m})^\top], \quad [C_{qq'}] = [m]^2.$$

It is a coordinate-basis covariance matrix, not an integral-kernel symbol. Under an invertible coordinate or quadrature reparameterization $m' = Sm$,

$$C' = SCS^\top, \quad t' = St, \quad W' = S^{-\top}WS^{-1},$$

so $t^\top Wm$ and $t^\top Wt$ are invariant. A complex representation replaces transpose by Hermitian adjoint. Thus self-adjointness and positivity mean $W_p = W_p^\dagger$ and $u^\dagger W_p u \geq 0$ in the declared coordinate representation.

The units satisfy $[t] = [m]/[a]$ and $[W] = [m]^{-2}$ for an exact covariance inverse; weaker weights may carry a common nonzero scale and different declared units because that scale cancels in n_p/d_p . In every case

$$[\hat{a}] = [m]/[t].$$

A point-source flux interpretation exists only if the template producer and CAL/BEAM lineage establish that exact amplitude convention. Otherwise the result is a declared shape amplitude.

8 Normative scientific definitions

8.1 Method, estimand, and fit boundary

SCI-FLT-MATCHED is an optimal matched-template amplitude estimator only under the exact criterion and premises in Section 9.3. It publishes a filtered amplitude map. At each fixed parent-pixel anchor it performs a fixed-template, fixed-anchor, one-parameter linear amplitude estimate. It does not select candidates, optimize position, fit morphology or background, deblend, infer source existence, or produce a catalog. It is neither posterior/Wiener sky reconstruction nor ordinary source-shaped convolution.

Each realization has the immutable scientific identity

```
(method_id, estimand, parent_id, parent_class, anchor_lattice, WCS/frame,
template_id, requested/effective/resolved/applied/realized generation,
weighting_object, exact_operator, realized_operator, numerical_profile,
response_family, support_family, units/calibration, covariance_scope,
NOI_lifecycle, product_bundle, validity, failure_policy, provenance).
```

A scientifically consequential change creates a new realization, generation, or separately named method. A representation-only change that exactly preserves the scientific object and supported query vocabulary does not change the estimand or scientific generation.

8.2 Parent classes and anchor lattice

Exactly one parent class is admitted per application: one immutable normalized ordinary-MAP observation bundle, or one immutable normalized ordinary-MAP coadd bundle with its exact contributing-observation membership. They are distinct applications. This contract asserts no filter/coadd commutation and owns no coaddition. JINC and other derived parents are not admitted. Numerical computability cannot repair missing MAP authority.

The base-v0.1 owner disposition places one output anchor at every exact parent ordinary-MAP pixel center. The output amplitude field preserves the parent’s WCS, frame, grid, shape, indexing, order, phase origin, and pixel-center convention, except that exact support rules may make anchors unavailable. No interpolation is part of anchor creation. Template translation at an anchor, including even/odd sampling and ties, follows the exact deterministic template object. Subpixel, oversampled, and independent output lattices are successors.

8.3 Template-response scientific object

Each application binds one immutable template-response scientific object, declared outside target, source, candidate, population, and NOI-member learning. It binds the unit-amplitude convention, source-domain reference, parent response, anchor/WCS relation, sampling, phase, support, CAL/BEAM lineage, validity, generation, and supported query vocabulary.

The same scientific object may be represented by exact materialization, an exact structured form, or exact lineage/query reconstruction. A representation-only change shall preserve every supported query and shall not alter the amplitude estimand. Template mismatch is not silently absorbed into normalization or parent covariance.

8.4 Five support roles

The following supports are distinct:

$\mathcal{S}_{\text{apply}}(p)$.

Exact parent payload rows with nonzero coefficients in $L_{p,g}$. Exact-zero coefficients neither require nor dereference their payloads.

$\mathcal{S}_{\text{template}}(p)$.

Exact rows and metadata required to define or query t_p .

$\mathcal{S}_{\text{state}}(g)$.

Complete population influencing learned covariance, spectra, reliability, window, rank, null, regularization, or other resolved state.

$\mathcal{S}_{\text{response}}(p, s)$.

Exact state, template, and operator support required by the declared response query.

$\mathcal{S}_{\text{store}}$.

Representation footprint. It has no scientific authority and may change losslessly.

An output anchor is admitted only when complete application support is present and valid. No partial-support estimator, edge renormalization, extension, imputation, background subtraction, learned taper, or signal-derived support belongs to this method. Numerical fill is allowed only after exact erosion proves zero application coefficient. Causal provenance retains $\mathcal{S}_{\text{state}}$ even when a fixed-state derivative touches only $\mathcal{S}_{\text{apply}}$.

8.5 Weighting and local covariance order

W_p is the selected frozen complete bilinear weight, exactly self-adjoint and positive semidefinite. A parent coefficient has only its selected-route role; its name, positivity, units, reciprocal, or diagonal placement does not make it covariance, precision, or inverse variance.

Local GLS first restricts the authorized parent covariance: $C_p = E_p C_{\text{parent}} E_p^T$, then inverts it only on the exact declared local estimable subspace. In general

$$(E_p C_{\text{parent}} E_p^T)^{-1} \neq E_p C_{\text{parent}}^{-1} E_p^T.$$

Using a global-precision subblock conditions on outside data and is a different method with a different influence domain, response, covariance, and edge rule.

8.6 Learn–Resolve–Apply state graph

State advances through three explicit operations:

Learn.

Derive candidate covariance, spectral/reliability information, window, rank, null, regularization, and diagnostics from one declared population, with source/target protection and state uncertainty recorded.

Resolve.

Select one owner-authorized SCI-FLT-MATCHED-A0-001 method and bind its parameters, support, scientific identity, numerical-conformance policy, failure disposition, and immutable generation. Externally declared state enters here without observational learning.

Apply.

Use the resolved state unchanged. No target, candidate, source, or NOI-member update occurs during Apply.

Requested, effective, learned-candidate, resolved, applied, realized, and published identities are distinct even when their values happen to agree. Observation and coadd states are separate. A compatible NOI member receives the identical resolved transformation, support, normalization, and failure rules. Any update creates a successor; per-member relearning is a different future method.

8.7 Fixed-state, full-procedure, realized, and reference response

The fixed-state response is $R_{\text{fixed}}(p, s) = L_{p, g_0} t_s^{\text{full}}$. It may be asymmetric, anisotropic, nonstationary, position-dependent, or nonlocal. A learned procedure has a separately typed full-procedure response obtained only by an authorized exact rerun or finite-difference definition that reruns Learn and Resolve after perturbation. The state-change record is retained separately. Until such authority exists, R_{FP} is unavailable; no FLT, FRUIT, SRC, or other consumer may call R_{fixed} the full-procedure response.

When the produced amplitude field uses a realized operator $\tilde{L}$, its actual response is $R_{\text{realized}} = \tilde{L}T$. The exact scientific comparator is $R_{\text{reference}} = LT$. These are not interchangeable.

8.8 Product roles and atomic lifecycle

The required atomic signal bundle contains the filtered amplitude field plus its quantity/unit, parent and anchor-lattice identity, template convention, resolved and realized operator/state identity, application validity, actual response reference, uncertainty availability, lifecycle, and complete provenance. Failure of any required member prevents a complete publication candidate.

Independently atomic qualified companions may carry authorized covariance, frozen-NOI uncertainty, response, or exact state/lineage. Every companion has its own scientific identity, support, population, unit, validity, lifecycle, and named-use policy. Optional failure does not change the signal estimand. No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is a public science product without a complete role and policy.

8.9 Lifecycle vocabulary

State	Meaning
not requested	No request exists. This is not disabled.
requested	A named method/product/use was requested; no admission is yet implied.
effective	Boundary resolution determined that the request applies to this parent and method identity.
disabled	An applicable requested route was explicitly disabled by its authorized state or policy.
unavailable	A required authority, input, option, support, or scientific premise is absent; no numerical value is asserted.
resolved	Method, state, parameters, support, policy, and failure disposition are immutable for the generation.
applied	The resolved operator was invoked on an admitted parent.
complete publication candidate	Every required member and predicate exists and the bundle is eligible for publication; this is not publication, validation, or readiness.
realized	The declared numerical operator produced the immutable product values and their actual response/covariance identities.

State	Meaning
failed	An authorized attempt began but a required operation or closure check failed.
not produced	Policy intentionally produced no artifact after a valid disposition; it is not zero or unavailable.
superseded	An immutable successor exists under an authorized policy; the predecessor is retained and never rewritten.

8.10 SCI-VAL named-use evaluation

SCI-VAL evaluates an owner-approved named-use profile. It retains four independent axes: request, applicability, eligibility, and realization. FLT owns the producer facts, dependency graph, named-use disposition, prescribed consumer action, and failure semantics. VAL registers and evaluates; it does not filter, publish, select an option, invent a missing fact, or perform observational/scientific validation. Each profile keeps separate verdicts for parent admission (PA), state admission (SA), signal publication (SP), conditional-covariance use (CU), NOI-companion use (NU), response-companion publication/use (RU), and FLT-to-FRUIT producer-envelope use (FH).

8.11 First-class FLT-to-FRUIT producer envelope

The FLT product contract preserves a first-class interface sufficient for a future FRUIT method to consume the filtered amplitude product without reconstructing undocumented FLT implementation state. At minimum it identifies the filtered quantity, template, response, valid support, units/calibration, method/state identity, uncertainty availability, and provenance. Additional FLT state required by an authorized FRUIT method remains exactly available or reconstructable through product lineage. FLT does not guess the future query vocabulary and defines no FRUIT model, recurrence, subtraction/add-back, learning, stopping, restart, selection, response, uncertainty, or interpretation.

8.12 Producer ownership and claim layers

MAP owns the immutable parent identity, support, response, covariance, beam, calibration, WCS, lifecycle, and provenance. The template producer owns the unit-amplitude response object. CAL/BEAM owns calibration and beam authority. FLT owns the exact transformation, resolved state, actual response/covariance consequences, products, lifecycle, and one-way producer envelope. NOI owns its realization population and empirical uncertainty meaning. FRUIT owns any future FRUIT science. No SRC boundary is created.

Scientific authority, implementation conformity, representation fidelity, observational/scientific validation, achieved performance, readiness, and production are separate claims. This draft establishes none beyond its own proposed scientific text.

9 Reference estimator, response, and uncertainty

9.1 Authoritative coordinate-basis operator

For every admitted anchor $p \in \mathcal{V}$,

$$m_p = E_p m, \quad t_p = E_p t_p^{\text{full}}, \quad (1)$$

$$n_p = t_p^{\text{T}} W_p m_p, \quad d_p = t_p^{\text{T}} W_p t_p, \quad (2)$$

$$\hat{a}_p = \frac{n_p}{d_p}, \quad L_{p,g} u = \frac{t_p^{\text{T}} W_p E_p u}{d_p}, \quad (3)$$

$$R_{\text{fixed}}(p, s) = L_{p,g} t_s^{\text{full}}. \quad (4)$$

$W_p = W_p^{\dagger}$ and $u^{\dagger} W_p u \geq 0$. The scientific normalization d_p is finite, exactly real, and strictly positive. Failure of one of these exact predicates is typed unavailability or attempt failure, never amplitude zero. Ordinary floating-point evidence for the exact identities belongs to the preregistered numerical-conformance profile.

9.2 Matching-template and general-sky relations

For the matching single-template model

$$m = A t_s^{\text{full}} + n, \quad \text{E}[n \mid g] = 0,$$

linearity gives

$$\text{E}[\hat{a}_p \mid g] = A R_{\text{fixed}}(p, s).$$

At $p = s$, exact normalization yields

$$R_{\text{fixed}}(p, p) = 1, \quad \text{E}[\hat{a}_p \mid g] = A.$$

For a general linear sky and deterministic residual,

$$m = \sum_s A_s t_s^{\text{full}} + b + n,$$

the exact fixed-state expectation is

$$\text{E}[\hat{a}_p \mid g] = \sum_s R_{\text{fixed}}(p, s) A_s + L_{p,g} b. \quad (5)$$

Unit diagonal response does not make off-diagonal response vanish. Neighboring or nonlocal templates may contribute, and a mean, offset, gradient, or other unmodeled deterministic b may bias the one-parameter amplitude. These terms are response and nuisance leakage, not source attribution or deblending.

9.3 Local restriction and constrained GLS theorem

For an authorized parent covariance, local restriction precedes inversion:

$$C_p = E_p C_{\text{parent}} E_p^{\text{T}}. \quad (6)$$

Let the columns of U_p be an exact orthonormal basis for the declared local estimable subspace and $P_p = U_p U_p^\top$ its exact projector. Define

$$C_{p,E} = U_p^\top C_p U_p, \quad W_p = U_p C_{p,E}^{-1} U_p^\top, \quad U_p^\top t_p \neq 0,$$

where $C_{p,E}$ is positive definite. This is not a claim that a restricted global inverse equals the inverse of the restricted covariance.

The competitor class contains exactly the linear estimates $\ell^\top m_p$ with: the identical parent support; the identical U_p , constraints, and fixed state; $\ell = P_p \ell$; and unit response $\ell^\top t_p = 1$ to the identical projected template. The matched GLS row is

$$\ell_* = \frac{W_p t_p}{t_p^\top W_p t_p}.$$

For every competitor,

$$\text{Var}(\ell^\top m_p \mid g) - \text{Var}(\ell_*^\top m_p \mid g) = (\ell - \ell_*)^\top C_p (\ell - \ell_*) \geq 0,$$

and

$$\text{Var}(\hat{a}_p \mid g) = d_p^{-1}.$$

This is the sole minimum-conditional-variance and d_p^{-1} variance route. It distinguishes covariance zero-variance modes, deliberately excluded or infinite-variance modes, weighting-null modes, and a projected template with no identifiable component. A shared numerical null label or an unconstrained Moore–Penrose operation does not establish the theorem.

9.4 Fixed-state and full-procedure response

For frozen state g_0 ,

$$R_{\text{fixed}}(p, s) = L_{p,g_0} t_s^{\text{full}}.$$

If state was learned from the parent, a full-procedure response exists only under separately authorized rerun authority. With

$$g_\epsilon = \text{Resolve}(\text{Learn}(m + \epsilon t_s^{\text{full}})),$$

it is the declared limit or predeclared finite difference

$$R_{\text{FP}}(p, s) = \lim_{\epsilon \rightarrow 0} \frac{\hat{a}_p(m + \epsilon t_s^{\text{full}}; g_\epsilon) - \hat{a}_p(m; g_0)}{\epsilon}, \quad (7)$$

or the exact finite- ϵ analogue with its step, sidedness, and domain fixed before evaluation. The state-change record is separate. Without this authority R_{FP} is unavailable.

9.5 Realized and reference response and covariance

Let L be the exact selected scientific operator and $\tilde{L}$ the operator that actually produced a field. Then

$$R_{\text{reference}} = LT, \quad C_{\text{reference}} = LC_{\text{parent}} L^\top, \quad (8)$$

$$R_{\text{realized}} = \tilde{L}T, \quad C_{\text{realized}} = \tilde{L}C_{\text{parent}} \tilde{L}^\top. \quad (9)$$

An amplitude field generated by $\tilde{L}$ shall carry the realized pair, never the reference pair as though it described the actual field. Either covariance exists only when matching C_{parent} authority exists. A diagonal representation does not assert independence, and omitted entries are unknown.

9.6 Engineering approximation observables

The following are comparison observables, not owner-selected scientific thresholds:

$$e_d(p) = \frac{|\tilde{d}_p - d_p|}{d_p}, \quad e_a(p) = |R_{\text{realized}}(p, p) - 1|, \quad (10)$$

$$e_R = \|P_{\text{out}}(R_{\text{realized}} - R_{\text{reference}})P_{\text{anc}}^{\text{T}}\|, \quad e_L = \|P_{\text{out}}(\tilde{L} - L)P_{\text{in}}^{\text{T}}\|, \quad (11)$$

$$e_C = \|P_{\text{out}}(C_{\text{realized}} - C_{\text{reference}})P_{\text{out}}^{\text{T}}\|. \quad (12)$$

One preregistered engineering numerical-conformance profile binds norms, scales, domains, projections, coverage, floating-point environment, exact subspace projectors, and acceptance rules before evaluation. No package-wide constant is scientific authority. If an authorized operator intentionally changes scientific response, support, null space, or estimand, it receives a separate method/realization identity and publishes that discrepancy.

9.7 Uncertainty budget and non-combinability

ID	Contribution	Contract treatment
U1	Parent stochastic covariance	Propagate with the operator that actually produced the field, for the exact authorized population and conditioning.
U2	Calibration/template scale	Preserve joint CAL/BEAM and parent-template authority; missing cross-covariance is unavailable, not zero.
U3	Learned-state uncertainty	Base products condition on frozen state. Marginalization over Learn-Resolve needs separate authority.
U4	Numerical or intentional operator discrepancy	Ordinary numerical error belongs to the preregistered engineering profile. Intentional scientific discrepancy changes method/realization identity.
U5	Covariance scope/representation	Declare complete, projected, or unavailable scope separately from exact representation.
U6	Frozen-NOI empirical uncertainty	A qualified companion with its own population and finite-sample meaning; it is not automatically parent covariance or physical noise.
U7	Template mismatch/unmodeled systematics	Outside the base matching- template covariance and separately reported.

No scalar total uncertainty combines these terms unless every term, cross-term, conditioning relation, unit, and population is authorized. Absence is unavailable, never zero.

10 Assumptions and validity domain

The unbiasedness, response, optimality, and covariance statements are valid only on the intersection of the following declared predicates. Every product shall identify which predicates were evaluated, by whose authority, and with what immutable evidence.

ID	Assumption or predicate
ASM-001	The parent is exactly one immutable normalized ordinary-MAP observation or coadd bundle with authoritative identity, generation, grouping, WCS/frame, support, units, calibration provenance, and coadd membership when applicable.
ASM-002	Parent, template, physical amplitude, filtered amplitude, and coordinate-basis covariance have declared real codomains. The parent and output anchor lattices have the owner-disposed common WCS/frame/grid/indexing relation. Coordinate or quadrature changes transform m, t, C, W according to Section 7.3; a complex representation preserves conjugacy and uses the Hermitian adjoint.
ASM-003	t_p is the one immutable template-response scientific object for the application and expresses expected parent response per unit declared amplitude, with fixed scale, anchor, phase, validity, lineage, and query vocabulary in an exact representation.
ASM-004	Complete $\mathcal{S}_{\text{apply}}(p)$ is present and valid. No admitted output depends on fill, extension, imputation, partial support, edge renormalization, adaptive background, learned taper, or signal-derived support. Template, state, response, and storage support remain distinct.
ASM-005	W_p is the exact frozen, self-adjoint, positive-semidefinite complete coordinate-basis bilinear weight named by the selected SCI-FLT-MATCHED-A0-001 alternative, including population, units, rank, null, regularization, window, support, and edge rules.
ASM-006	d_p is finite, exactly real, and strictly positive.
ASM-007	For amplitude unbiasedness, the admitted noise has zero conditional mean on the complete support given the frozen state and does not induce target-dependent changes in the template or operator.
ASM-008	For GLS optimality, C_{parent} is the exact authorized covariance for the population and conditioning; C_p is restricted before inversion; $C_{p,E}$ is positive definite on the exact estimable subspace; the projected template is identifiable; and the competitor class is identical.
ASM-009	For covariance, the parent law, operator that actually produced the field, response, support, units, and conditioning state refer to the same realization and generation; reference quantities remain separate.
ASM-010	State follows the declared Learn-Resolve-Apply graph and is frozen before Apply. It is neither target-updated nor NOI-member relearned.
ASM-011	A compatible NOI ensemble member receives the identical state, operator, support, normalization, numerical profile, and failure policy as the signal parent.
ASM-012	A realized numerical operator is assessed under one preregistered engineering numerical-conformance profile. An intentional scientific discrepancy has separate method/realization identity.
ASM-013	Covariance scope and representation, response query vocabulary and representation, and state query vocabulary and representation satisfy the selected SCI-FLT-MATCHED-A0-003-SCI-FLT-MATCHED-A0-005 lifecycle rules.
ASM-014	Template and parent calibration lineages are compatible; any claimed point-source flux meaning or calibration covariance has explicit CAL/BEAM authority.

ID	Assumption or predicate
ASM-015	A method option, named-use policy, and lifecycle route are explicitly selected; there is no automatic selector, fallback, or silent substitution.

10.1 Validity table

Dimension	Admission evidence	Invalid or missing case	Required outcome
Parent identity	Exact bundle identity, class, generation, grouping, WCS/frame, support, unit, quantity, calibration lineage, and coadd membership	Missing, mutable, derived, or mixed parent	unavailable before application; no FLT repair or coaddition.
Application class	Observation or coadd declared independently	Assumed commutation or borrowed state between classes	unavailable; require a distinct realization.
Output anchors	One anchor at each exact parent pixel center with identical parent WCS, frame, grid, shape, index, order, and pixel-center convention	Interpolated, subpixel, oversampled, independently sampled, or ambiguous phase/tie rule	unavailable under v0.1; require a separately named successor.
Template	Exact response object per unit amplitude with scale, phase, anchor, sampling, lineage, validity, and query vocabulary	Learned from target/member, mismatched chart, ambiguous scale, lossy representation, or missing lineage	unavailable; no inferred template.
Application support	Complete nonzero-coefficient payload closure and erosion proof, separately typed from state, template, response, and storage support	Missing, invalid, filled, extended, imputed, or partially supported dependency	Mark affected anchors unavailable; no partial estimate.
Weighting state	Selected SCI-FLT-MATCHED-A0-001, population, self-adjoint PSD operator, rank/null, regularization, coefficients, window and edge rules bound at Resolve	Unknown, unselected, incompatible, or changing W_p	Fail closed; no fallback.
Normalization	Finite, exactly real, strictly positive d_p	NaN, infinite, complex, zero, negative, or null-confounded	Location unavailable or attempted bundle failed; never write zero.
Numerical realization	One preregistered engineering conformance profile with the exact comparator, projectors, norms, scales, domain, and acceptance rules	Post-hoc profile, uncovered modes, support/subspace mismatch, or failed rule	failed for that profile; a scientific operator change requires distinct identity.

Dimension	Admission evidence	Invalid or missing case	Required outcome
Response	Fixed, full-procedure, realized, and reference types kept distinct, with selected exact domain/query/representation contract	Undocumented kernel, assumed invariance, strengthened fixed-state meaning, or unreconstructable response	Response-dependent product or handoff unavailable.
Units and calibration	$[\hat{a}] = [m]/[t]$ and compatible joint lineage	Unjustified flux, beam, solid angle, DC, extended-source, or covariance meaning	Forbid the claim; required named use is unavailable.
Covariance	Matching authorized C_{parent} , actual producing operator, declared scientific scope, and exact selected representation	Missing population/conditioning, omitted correlations treated as zero, or d_p^{-1} used without constrained local GLS	Covariance use unavailable or companion failed; signal may remain complete publication candidate.
NOI parity	Identical frozen transformation for compatible members	Per-member relearning, different support, or mixed generation	NOI companion unavailable or failed; separate method required.
Lifecycle	Atomic membership, named-use policy, and immutable state	Mixed status, partial signal bundle, silent mutation, or overwritten predecessor	Bundle failed; retain predecessor and successor identities.
FLT to FRUIT	Selected SCI-FLT-MATCHED-A0-005, documented response/state lineage, and handoff profile	Requires undocumented implementation state or supplies FRUIT science	Handoff unavailable; signal product status is unchanged.

The valid output domain $\mathcal{V}$ is the intersection of all location-level predicates above. Bundle-level validity additionally requires all atomic identity and lifecycle members. Consumer-specific validity is never broader than producer validity and may be narrower under a named-use policy.

11 Normative requirements

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-001	The method identity shall be SCI-FLT-MATCHED, titled “Optimal matched-template map filtering,” subject to the exact qualification in Section 9.3.
SCI-FLT-MATCHED-REQ-002	The sole signal estimand shall be the fixed-template amplitude at each admitted anchor; the public signal role is a filtered amplitude map.
SCI-FLT-MATCHED-REQ-003	Each application shall bind exactly one immutable normalized ordinary-MAP observation parent or one immutable normalized ordinary-MAP coadd parent. JINC and other derived parents are excluded.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-004	Observation and coadd applications shall remain distinct; no commutation, equivalence, state sharing, or FLT-owned coaddition is asserted.
SCI-FLT-MATCHED-REQ-005	Each application shall bind one immutable template-response scientific object per unit amplitude. Exact materialized, structured, and lineage/query representations are admissible only when scientifically equivalent.
SCI-FLT-MATCHED-REQ-006	The template shall not be learned or updated from a target, source, candidate, population, or NOI realization member.
SCI-FLT-MATCHED-REQ-007	A realization shall bind collision-free $p/q/s/W_p$ notation, coordinate-basis algebra, domain, real codomains, WCS/frame, indexing, units, support families, boundaries, estimable subspace, rank/null, regularization, validity, and any complex-transform adjoint convention.
SCI-FLT-MATCHED-REQ-008	The authoritative estimator shall be exactly $n_p = t_p^T W_p m_p$, $d_p = t_p^T W_p t_p$, and $\hat{a}_p = n_p/d_p$, with every scientific measure factor inside W_p .
SCI-FLT-MATCHED-REQ-009	Every admitted W_p shall be exactly self-adjoint and positive semidefinite and every admitted d_p finite, exactly real, and strictly positive. Numerical tolerances do not alter these scientific predicates.
SCI-FLT-MATCHED-REQ-010	Complete $\mathcal{S}_{\text{apply}}(p)$ shall be required. Partial support, renormalization, extension, imputation, background subtraction, learned taper, and signal-derived support are excluded.
SCI-FLT-MATCHED-REQ-011	Fixed-state, full-procedure, realized, and reference response shall remain separately typed. Required response queries shall be exactly available or reconstructable; a reduced kernel requires proved invariances and measure.
SCI-FLT-MATCHED-REQ-012	“Optimal” and d_p^{-1} variance require the constrained local inverse-covariance GLS theorem: restriction before inversion, a positive-definite estimable covariance, identifiable projected template, identical constraints/support/state, and the exact competitor class.
SCI-FLT-MATCHED-REQ-013	Every realization shall identify one owner-selected SCI-FLT-MATCHED-A0-001 weighting alternative separately for its parent class. No option in this draft is a default.
SCI-FLT-MATCHED-REQ-014	A parent coefficient shall retain only its declared route-specific role; its name, sign, units, reciprocal, or diagonal form does not imply covariance, precision, inverse variance, or independence.
SCI-FLT-MATCHED-REQ-015	State shall follow Learn–Resolve–Apply. Requested, effective, candidate, resolved, applied, realized, and published generations remain distinct; Apply uses immutable state and performs no target/member update.
SCI-FLT-MATCHED-REQ-016	Observation and coadd state generations shall remain separate and shall not be substituted for one another.
SCI-FLT-MATCHED-REQ-017	Every compatible NOI member shall receive the identical resolved state, transformation, application support, normalization, numerical profile, and failure rules as the corresponding signal realization.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-018	An NOI-informed update shall create an immutable successor; per-member relearning is a different, unavailable-in-v0.1 method.
SCI-FLT-MATCHED-REQ-019	The method shall have no automatic selector or fallback. A requested unavailable route fails closed without silent substitution.
SCI-FLT-MATCHED-REQ-020	When matching parent-covariance authority exists, a produced field shall carry $C_{\text{realized}} = \tilde{L}C_{\text{parent}}\tilde{L}^T$; $C_{\text{reference}} = LC_{\text{parent}}L^T$ remains a separate comparator.
SCI-FLT-MATCHED-REQ-021	d_p^{-1} shall be called marginal conditional variance only under every constrained local GLS premise; otherwise d_p is normalization.
SCI-FLT-MATCHED-REQ-022	Uncertainty availability and U1–U7 shall be truthful and role-separated; missing covariance, cross-covariance, or entries are unavailable rather than zero or independent.
SCI-FLT-MATCHED-REQ-023	The public signal product shall be the role-complete atomic bundle defined in Section 8.8.
SCI-FLT-MATCHED-REQ-024	Failure of any required signal member shall prevent a complete publication candidate; a partial bundle shall not be published as complete.
SCI-FLT-MATCHED-REQ-025	Covariance, frozen-NOI uncertainty, response, and state/lineage companions shall be independently atomic and qualified by exact identity, population, unit, support, validity, lifecycle, and named-use policy.
SCI-FLT-MATCHED-REQ-026	The lifecycle states in Section 8.9 shall remain distinct; in particular not requested is not disabled , and a complete publication candidate is not publication, validation, or readiness.
SCI-FLT-MATCHED-REQ-027	No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is public without a complete science-product role and policy.
SCI-FLT-MATCHED-REQ-028	The product contract shall preserve a first-class FLT-to-FRUIT producer envelope identifying filtered quantity, template, response, valid support, units/calibration, method/state, uncertainty availability, and provenance without undocumented implementation state.
SCI-FLT-MATCHED-REQ-029	No FLT artifact or profile shall define FRUIT source modeling, subtraction/add-back, recurrence, learning, stopping, restart, selection, response, uncertainty, or interpretation.
SCI-FLT-MATCHED-REQ-030	The method performs fixed-template, fixed-anchor, one-parameter linear amplitude estimation. Candidate selection, position/morphology/ background fitting, deblending, existence inference, catalogs, significance, completeness, purity, and SRC ownership remain outside scope.
SCI-FLT-MATCHED-REQ-031	Posterior/Wiener sky reconstruction, ordinary convolution, data-thresholded destriping, adaptive edge/background methods, and the historical high-pass/delta case remain separate methods. SCI-FLT-FIXED remains a separate protected package.
SCI-FLT-MATCHED-REQ-032	Every affected AO route shall receive its exact required owner or engineering disposition before dependent scientific use is available. An AO selection cannot manufacture an unavailable parent, template, or authority.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-033	The scientific method is the exact selected n_p/d_p operator. Ordinary finite precision is evaluated under one preregistered engineering numerical-conformance profile. A scientifically different operator receives a separate method/realization identity and retains its discrepancy.
SCI-FLT-MATCHED-REQ-034	Scientific authority, implementation conformity, representation fidelity, observational validation, achieved performance, readiness, and production shall be separate claims with separate evidence and owners.
SCI-FLT-MATCHED-REQ-035	SCI-VAL shall evaluate an owner-approved named-use profile on separate request, applicability, eligibility, and realization axes. It shall not perform filtering, publication, or observational/scientific validation.
SCI-FLT-MATCHED-REQ-036	Parent and template calibration dependence shall remain joint. Point-source flux, matched-filter beam/solid angle, extended-source fidelity, and calibration covariance require exact separate authority.
SCI-FLT-MATCHED-REQ-037	A coadd application shall retain exact contributing-observation membership and parent provenance without asserting filter/coadd commutation.
SCI-FLT-MATCHED-REQ-038	Every product shall bind exact MAP, template, NOI, and FLT-to-FRUIT boundary records, the full realization tuple, immutable provenance, supported query/reconstruction contract, and a route-status record.
SCI-FLT-MATCHED-REQ-039	This Stage B draft and every unselected option shall make no implementation, conformity, covariance/response fidelity, observational validation, performance, readiness, production, scientific-freeze, or Unity claim.
SCI-FLT-MATCHED-REQ-040	Base v0.1 shall evaluate one anchor at each exact parent pixel center and preserve parent WCS/frame/grid/shape/index/order/pixel-center conventions. No interpolation is admitted; other lattices are successors.
SCI-FLT-MATCHED-REQ-041	Local covariance shall be $C_p = E_p C_{\text{parent}} E_p^T$ and local GLS shall invert only its exact estimable restriction. A global-precision sub-block is a separate conditional method.
SCI-FLT-MATCHED-REQ-042	$\mathcal{S}_{\text{apply}}$, $\mathcal{S}_{\text{template}}$, $\mathcal{S}_{\text{state}}$, $\mathcal{S}_{\text{response}}$, and $\mathcal{S}_{\text{store}}$ shall be separately recorded; exact-zero application coefficients do not dereference parent payload.
SCI-FLT-MATCHED-REQ-043	The general-sky expectation shall retain all off-diagonal template response and $L_{p,g}b$ nuisance leakage. Unit diagonal response shall not imply source attribution, isolation, or absence of bias.
SCI-FLT-MATCHED-REQ-044	When state is parent-learned, full-procedure response requires an authorized Learn-Resolve rerun or predeclared finite difference and a separate state-change record; otherwise it is unavailable.
SCI-FLT-MATCHED-REQ-045	Any amplitude field generated with $\tilde{L}$ shall carry $R_{\text{realized}} = \tilde{L}T$ and any authorized C_{realized} , not reference quantities mislabeled as actual.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-046	Covariance scientific scope shall be selected independently as complete, named projected, or unavailable; exact representation shall be selected independently and shall not change covariance identity.
SCI-FLT-MATCHED-REQ-047	Exact immutable state and its required query vocabulary shall be reproducible across full, compact, or lineage representation without changing the estimator or signal generation.
SCI-FLT-MATCHED-REQ-048	Response domain, query vocabulary, validity, and consumer scope shall be defined before choosing materialized, structured, or lineage representation; FLT shall not guess future FRUIT requirements.
SCI-FLT-MATCHED-REQ-049	PA, SA, SP, CU, NU, RU, and FH verdict meanings and their dependency graph are normative. Profile count/layer/vector layout is lossless representation only; a scalar verdict shall not drive scientific action.
SCI-FLT-MATCHED-REQ-050	Covariance zero-variance modes, excluded or infinite-variance modes, weighting-null modes, and template-unidentifiable modes shall remain distinct exact subspace classifications.

12 Falsifiable scientific predictions

Each prediction is conditional on the cited assumptions and is falsified by a conforming counterexample in its declared domain. Failure authorizes no fallback.

Prediction	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-001	An exact unit-amplitude matching template at admitted p gives $R_{\text{fixed}}(p, p) = 1$.	003–006
SCI-FLT-MATCHED-PRED-002	At fixed state, a noiseless matching template scaled by A gives exactly $\hat{a}_p = A$.	003–006
SCI-FLT-MATCHED-PRED-003	With zero conditional-mean noise and a matching template, $E[\hat{a}_p g] = A$.	003–007
SCI-FLT-MATCHED-PRED-004	An anchor missing any nonzero application-support dependency is never published as a numerical amplitude.	004,006
SCI-FLT-MATCHED-PRED-005	Nonfinite, nonreal, zero, or negative d_p never produces zero amplitude as a substitute.	005–006
SCI-FLT-MATCHED-PRED-006	Observation and coadd applications retain distinct parent and state identities.	001,010
SCI-FLT-MATCHED-PRED-007	Every compatible NOI member has the identical fixed L , support, normalization, profile, and failure decisions as its signal realization.	010–011
SCI-FLT-MATCHED-PRED-008	For draws from the exact declared parent stochastic law with conditional covariance C_{parent} , empirical covariance of fixed-state transformed fields converges to $LC_{\text{parent}}L^T$ under the declared sampling theorem and uncertainty. An NOI ensemble supports this only when separate authority establishes the same target covariance.	008–011

Prediction	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-009	Without exact constrained local GLS premises, removing a d_p^{-1} variance label changes no amplitude value and removes the unsupported uncertainty claim.	005,008–009
SCI-FLT-MATCHED-PRED-010	A realized operator accepted under its preregistered engineering profile satisfies every predeclared numerical rule; this is conformance to the profile, not a scientific error budget or performance claim.	012–013
SCI-FLT-MATCHED-PRED-011	A consequential realization-field change creates a new identity and never mutates an existing product.	010, 013, 015
SCI-FLT-MATCHED-PRED-012	Failure of one required signal member prevents a complete publication candidate.	015
SCI-FLT-MATCHED-PRED-013	Failure of an optional companion leaves a complete signal bundle unchanged unless its named-use policy requires that companion.	009, 013, 015
SCI-FLT-MATCHED-PRED-014	Rescaling $t \mapsto ct$ gives $\hat{a} \mapsto \hat{a}/c$ while preserving $\hat{a}t$.	003,014
SCI-FLT-MATCHED-PRED-015	A one-kernel response representation fails where its required invariances do not hold.	002,013
SCI-FLT-MATCHED-PRED-016	Requesting an unavailable route never silently selects another route.	015
SCI-FLT-MATCHED-PRED-017	A signal bundle plus its declared response/state interface answers every supported FLT-side handoff query without undocumented state.	013,015
SCI-FLT-MATCHED-PRED-018	No conforming output alone is a detection, catalog entry, posterior sky, FRUIT result, performance claim, readiness decision, or production decision.	001–015
SCI-FLT-MATCHED-PRED-019	With two nonzero template amplitudes, the expectation at p equals $R(p, s_1)A_{s_1} + R(p, s_2)A_{s_2}$ plus any nuisance term.	003–009
SCI-FLT-MATCHED-PRED-020	A nonzero constant or gradient background changes $E[\hat{a}_p]$ by exactly $L_{p,g}b$ when that contraction is nonzero.	003–009
SCI-FLT-MATCHED-PRED-021	A nonzero off-diagonal $R_{\text{fixed}}(p, s)$ produces a response at p to a unit template at $s \neq p$ despite exact unit diagonal response.	003–009
SCI-FLT-MATCHED-PRED-022	A mismatched signal shape returns its response-weighted template amplitude and need not equal any unconstrained source property.	003–009
SCI-FLT-MATCHED-PRED-023	For parent-learned state, an authorized full-procedure response can differ from fixed-state response exactly through the recorded state change; without rerun authority the former remains unavailable.	010
SCI-FLT-MATCHED-PRED-024	If $\tilde{L} \neq L$, empirical covariance of draws transformed by $\tilde{L}$ converges to C_{realized} , not to $C_{\text{reference}}$, under the declared sampling theorem.	008–012

13 Unselected authored option sets

All alternatives below remain unselected. Every scientific weight is exact, self-adjoint, positive semidefinite, and used through exact normalization. Observation and coadd dispositions and generations remain separate.

13.1 SCI-FLT-MATCHED-A0-001: Weighting and noise authority

Each route binds the map chart, units, support, population, Learn/Resolve identity, exact subspaces, window/boundary action, regularization, coefficient roles, diagnostics, state uncertainty, generations, and failure disposition.

13.1.1 SCI-FLT-MATCHED-A0-001-A: exact constrained local inverse-covariance GLS

Science.

Restrict exact authorized C_{parent} with E_p , then use $W_p = U_p C_{p,E}^{-1} U_p^T$ on the declared estimable subspace with identifiable projected template. This alone supports the theorem in Section 9.3.

Failure and evidence.

Missing covariance authority, population or conditioning mismatch, nonpositive $C_{p,E}$, subspace mismatch, or unidentifiable template makes GLS/optimalty unavailable. Future assessment checks the exact theorem and the preregistered numerical profile separately.

13.1.2 SCI-FLT-MATCHED-A0-001-B: structured covariance-derived weighting

Science.

Resolve one exact self-adjoint PSD structured weight derived from a declared covariance family. The selected W_p is exact science even when it is not the exact covariance inverse. Normalization and unit response remain; optimality and d_p^{-1} variance do not.

Failure and evidence.

Unbound structure, population, modes, support, regularization, or covariance ancestry fails the route. Any comparison with an inverse covariance is an engineering/profile observable, not an implied claim.

13.1.3 SCI-FLT-MATCHED-A0-001-C: radially symmetrized field-power spectral weighting

Object.

Resolve a declared field population f_j and transform

$$F_j(k) = \alpha \sum_q H(q) f_j(q) e^{-2\pi i k \cdot r_q} \mu_q, \quad \bar{P}_b = \frac{\sum_{j,k \in B_b} w_{jk} |F_j(k)|^2}{\sum_{j,k \in B_b} w_{jk}}.$$

The source may be an independently authorized noise population or a declared parent-derived field. Its name and deterministic-residual imprint are retained.

Exact conventions.

Each radial bin is half-open $B_b = [\rho_b, \rho_{b+1})$ except the declared final-boundary rule; boundary ties use one deterministic assignment. Member/mode weights are finite and nonnegative and their denominator is positive. Phase, normalization, units, conjugate modes, multiplicities, averaging order, WCS metric, window, and support are exact.

Weight and modes.

On admitted modes $w_{\text{spec}}(k) = (\bar{P}_{b(k)} + \lambda_b)^{-1}$ with $\lambda_b \geq 0$ in the units of $\bar{P}_b$. Regularized modes

and excluded-null modes are distinct. A sparse bin is never interpolated or borrowed from neighbors without a separately authored rule.

Consequences.

Radialization deliberately loses angular information and implies no noise, covariance, stationarity, isotropy, or optimality claim. d_p is normalization. Matching covariance, if any, requires independent authority.

Failure and evidence.

Invalid weights/denominator, nonfinite or negative power, unbound ties/multiplicity, missing field provenance, incomplete support, or hidden borrowing fails the route. Future evidence reports bin membership, source imprint, anisotropy/dispersion, leakage, regularization, response, and NOI parity without promoting them to scientific thresholds.

13.1.4 SCI-FLT-MATCHED-A0-001-D: declared weaker positive-semidefinite weighting

Science.

Resolve any other exact self-adjoint PSD map or transform-domain weight with declared purpose, population, units, support, rank/null, regularization, and lineage, but no covariance ancestry claim.

Failure and evidence.

Non-self-adjointness, indefiniteness, hidden coefficients, state drift, incomplete support, or invalid d_p fails the route. No optimality or variance meaning follows.

13.2 SCI-FLT-MATCHED-A0-002: Operator realization and numerical-conformance class

No numerical constant in this family is scientific authority. Exactly one engineering profile is preregistered per realized numerical identity.

13.2.1 SCI-FLT-MATCHED-A0-002-A: numerical realization of the exact operator

The scientific operator, response, support, subspaces, and failure semantics are exact. The preregistered engineering profile binds floating-point format, algorithms, environment, exact comparator, domains, norms, scales, projectors, coverage, and acceptance rules before results are seen. Profile pass is only a numerical-conformance statement.

13.2.2 SCI-FLT-MATCHED-A0-002-B: intentionally distinct scientific operator

This route is admissible only through separate owner authority that states the affected quantity, estimand, response/support/null discrepancy, scientific error budget, norm/domain, calibration consequence, consumers, and rationale. It creates a distinct method or realization identity; it is not ordinary finite precision and cannot borrow the exact operator's response/covariance.

13.2.3 SCI-FLT-MATCHED-A0-002-C: numerical route unavailable

If no preregistered profile, exact comparator, domain/coverage, or required projector exists, no numerical-conformance claim is available. No threshold, iteration count, percentile, or post-hoc test supplies a fallback.

13.3 SCI-FLT-MATCHED-A0-003: Conditional-covariance scope and representation

This family requires one scientific-scope disposition from A–C and, for an available scope, one exact-representation disposition from D–E. Equivalent exact representations retain one covariance identity.

13.3.1 SCI-FLT-MATCHED-A0-003-A: complete scientific covariance scope

Every authorized entry and correlation of the actual produced field is in scope on the full valid output domain, with exact population, conditioning, units, rank/null, calibration terms, response, and generation.

13.3.2 SCI-FLT-MATCHED-A0-003-B: named projected scientific covariance scope

Only $P_c C_{\text{realized}} P_c^\top$ for a named consumer projection is in scope. The projection basis, measure, rank, validity, and consumer are exact; all outside correlations remain unknown.

13.3.3 SCI-FLT-MATCHED-A0-003-C: covariance unavailable

Missing parent-covariance or cross-term authority is typed unavailability. The signal may remain complete, but no variance, standardization, significance, draw, independence, or covariance-dependent use follows.

13.3.4 SCI-FLT-MATCHED-A0-003-D: exact resident covariance representation

Represent the selected scope by an explicit array or exact structured object. The representation type and query vocabulary are declared; every supported entry/quadratic form is exact, and no representation-only change alters the scientific covariance identity.

13.3.5 SCI-FLT-MATCHED-A0-003-E: exact lineage/on-demand covariance representation

Persist exact immutable lineage sufficient to reconstruct every supported query without undocumented state. Each result is immutable and carries the same covariance identity. Missing lineage or unsupported queries are unavailable, never zero.

13.4 SCI-FLT-MATCHED-A0-004: Immutable-state representation

The scientific invariant for every alternative is exact reproducibility of the resolved immutable state and its required query vocabulary. Representation alone does not change the estimator or signal generation.

13.4.1 SCI-FLT-MATCHED-A0-004-A: full materialization

Persist every resolved state element with exact indexing, units, population, generation, and canonical identity. Reload shall reproduce every supported operator/state query exactly under the numerical profile.

13.4.2 SCI-FLT-MATCHED-A0-004-B: compact exact state

Persist an exact factorization or sufficient-state representation plus complete query semantics. Discarded information is allowed only when proved irrelevant to every supported query. Unsupported queries are unavailable.

13.4.3 SCI-FLT-MATCHED-A0-004-C: exact lineage reconstruction

Persist immutable inputs, Learn/Resolve algorithms and versions, environment facts, and identity needed for deterministic reconstruction. Missing or mutable lineage fails dependent routes; relearning is not a fallback.

13.5 SCI-FLT-MATCHED-A0-005: Response science and exact representation

Before representation, every route binds fixed/FP/realized/reference type, anchor domains, index/measure, query vocabulary, validity, units/calibration, state/operator/template generation, consumer scope, retention, and failure. FLT does not guess a future FRUIT query.

13.5.1 SCI-FLT-MATCHED-A0-005-A: full response materialization

Materialize every response entry in the authorized domain with exact index maps and identity. A matrix does not by itself authorize a beam, moment, integral, or other reduced interpretation.

13.5.2 SCI-FLT-MATCHED-A0-005-B: exact structured response

Persist an exact factorization or mathematical object that answers the entire declared query vocabulary without unproved invariance. Unsupported queries are typed unavailable.

13.5.3 SCI-FLT-MATCHED-A0-005-C: exact lineage/on-demand response

Persist complete immutable lineage needed to reconstruct supported response queries without undocumented state. Query results are immutable; failure does not substitute a reduced kernel.

13.6 SCI-FLT-MATCHED-A0-006: Role-separated SCI-VAL verdict representation

The normative policy is one dependency graph over seven separate verdicts: PA, SA, SP, CU, NU, RU, and FH. Each retains request, applicability, eligibility, realization, producer facts, policy, action, lifecycle, evidence, and version. The alternatives are lossless representations of the same policy.

13.6.1 SCI-FLT-MATCHED-A0-006-A: seven separate records

Persist one immutable record per role. A view may list all verdicts but has no actionable scalar aggregate.

13.6.2 SCI-FLT-MATCHED-A0-006-B: grouped layered records

Group roles into declared layers only while retaining seven independently addressable subverdicts and the full dependency graph. A layer summary cannot replace the subverdict vector.

13.6.3 SCI-FLT-MATCHED-A0-006-C: one seven-role verdict vector

Persist one composite with seven separately addressable complete records. Consumers request a named component. A scalar may be human-readable only and shall never drive action.

13.7 Option closure

Until exact dispositions exist, affected weighting, numerical-conformance, covariance, state-query, response-query, SCI-VAL, and handoff routes remain unavailable. Selection establishes no implementation conformity, response or covariance fidelity, observational validation, performance, readiness, production, freeze, or Unity status.

14 Edge cases, exclusions, and failure routing

Condition	Scientific meaning	Required handling
Empty or incomplete application support	No defined reference estimand at the affected anchor	Mark unavailable before evaluation; no padding, extension, imputation, partial estimate, or renormalization.
Numerical fill near an edge	Merely a computational placeholder	Admit an output only after exact erosion proves zero dependency on every filled value.
$d_p = 0$, $d_p < 0$, nonfinite d_p , or complex d_p	Template is null, weighting is invalid, or numerical identity failed	Typed unavailability or attempt failure; never publish zero amplitude, infinite variance, or fallback.
Template lies partly in $\mathcal{N}(W_p)$	Some declared template modes are unestimable	Allowed only if the selected option explicitly defines the same estimable subspace and $d_p > 0$; record response loss and null space. Otherwise unavailable.
Indefinite or non-self-adjoint W_p	Weight is outside the admitted scientific bilinear family semantics	Fail selected SCI-FLT-MATCHED-A0-001 closure unless an exact distinct operator and inner product were separately authored; no silent symmetrization.
Covariance variance mode	zero-Parent law declares deterministic variation in that direction	Keep separate from excluded, weighting-null, and template-unidentifiable modes; apply only the exact constrained theorem.

Condition	Scientific meaning	Required handling
Excluded or infinite-variance mode	Policy deliberately removes a direction from the estimable subspace	Record the exact projector and response loss; do not call it a covariance zero mode.
Weighting-null mode	W_p assigns zero action to a direction	Record the weighting null space; no covariance meaning follows without separate authority.
Projected template is zero	The requested amplitude is unidentifiable in the declared subspace	Anchor unavailable; no Moore–Penrose or normalization fallback.
Zero, negative, or non-finite field-power bin	No finite declared reciprocal weighting for that mode/bin	Apply only owner-declared regularization within the frozen state; otherwise exclude as a declared null mode or make route unavailable. Never infer infinite precision.
Sparse radial bin or large angular scatter	Radially averaged PSD lacks declared information or radial sufficiency	Fail owner-set effective-sample, anisotropy, or leakage bounds; no neighboring-bin interpolation unless separately authored.
Unknown covariance entry	Scientific knowledge is absent	Preserve unknown; do not encode zero, diagonal independence, or a usable inverse.
Available W but unavailable C_{parent}	Weighting exists without covariance authority	Amplitude filtering may proceed under a selected weaker option; optimality, d_p^{-1} variance, and conditional covariance are unavailable.
Available C_{parent} with mismatched population/state	Covariance does not describe this realization	Covariance route unavailable; no rescaling or population substitution.
Template rescaled by $c \neq 0$	Amplitude convention changes	Create a new template/realization identity; $\hat{a}$ rescales by $1/c$ and units/lineage must follow.
Template phase or anchor changed	Scientific response convention changes	New template and realization; never treat as a numeric implementation detail.
Template mismatch to a real signal	Base matching-template estimand does not equal an unconstrained source property	Report only response-weighted shape amplitude; no bias correction, morphology, or new uncertainty without separate authority.
Parent beam or calibration exists	Provenance is present, not inherited matched-filter meaning	Preserve lineage; require exact authority before flux, matched beam/solid angle, extended-source, or calibration-covariance claim.

Condition	Scientific meaning	Required handling
Coadd membership changes	Parent identity changes	New coadd parent, state, realization, response, and products; do not reuse observation/coadd state by implication.
NOI member suggests a new state	Per-member relearning would change the ensemble method	Apply the frozen state or mark member incompatible; a learned update creates a successor generation outside the current ensemble.
Approximation misses one null or support mode	Estimand/validity differs even if aggregate error is small	Fail the preregistered profile; if intentional, assign a distinct scientific identity. No percentile hides the disagreement.
Iteration/tail cap reached	Work limit, not convergence evidence	Accept only if every selected quantitative envelope bound independently passes; otherwise fail.
Optional companion fails after signal completion	Companion atomicity failed, not necessarily signal science	Mark companion failed; preserve signal complete publication candidate unless exact named-use policy made the companion required.
Required signal member fails	Role-complete signal does not exist	Fail the entire signal bundle; do not publish a partial bundle as unavailable pixels inside an otherwise complete product.
Successor state or option is approved	Scientific identity advances	Create immutable successor generation and mark predecessor superseded only by policy; never mutate predecessor facts.
Unselected or unavailable option requested	No authorized route exists	Fail closed and name the missing owner disposition; no automatic selector or fallback.
Response query is unsupported	Scientific interface does not define that quantity	Return typed unavailability; never approximate with a kernel, radial profile, parent beam, or zero.
Fixed-state response requested as full-procedure response	State-learning dependence has not been rerun	Return full-procedure response unavailable and retain the fixed-state type; no consumer may strengthen it.
Reference response or covariance attached to a realized field	Comparator is being mislabeled as the actual product consequence	Fail the companion or bundle policy; attach R_{realized} and C_{realized} when authorized.
FLT to FRUIT consumer asks for FRUIT behavior	Request crosses package authority	Withhold the behavior and require a separate FRUIT contract; the FLT signal and handoff facts remain unchanged.

Condition	Scientific meaning	Required handling
Filtered-map peak is examined	A numerical feature is not an authorized source object	No detection, candidate, significance, completeness, purity, fit, morphology, catalog, or SRC claim follows.

14.1 Explicit method exclusions

Posterior/Wiener sky reconstruction, ordinary convolution, source or candidate learning, data-thresholded destriping, automatic selectors/fallbacks, adaptive edge/background behavior, the historical high-pass/delta case, partial-support estimation, FLT-owned coaddition, source analysis, and FRUIT algorithms are not degenerate settings of this contract. They require separate scientific contracts if later commissioned.

SCI-FLT-FIXED remains a separate protected package. This contract neither selects, modifies, supersedes, nor supplies scientific authority for that method.

The historical compatibility term “Wiener filter” may refer to this method only when accompanied by the exact SCI-FLT-MATCHED identity and may not imply a sky prior, posterior estimand, or Wiener reconstruction.

14.2 Nonclaim closure

Passing a scientific prediction or an engineering conformance test establishes only that named claim for the exact immutable realization. Validation-profile pass is not achieved performance. Achieved performance is not readiness. Readiness is not production. None of those layers can select an authored option, repair missing scientific authority, or broaden this validity domain.

Part III

Owner-disposition guide

15 What remains for the scientific owner

This draft refactors, but does not select, six decision families. The owner must dispose the remaining scientific questions, while one engineering numerical profile is separately preregistered, before dependent routes become available. The decision ledger records the exact open questions.

Family	Decision	Route unavailable before disposition
SCI-FLT-MATCHED-A0-001	Weighting authority for observation and coadd separately; all population, coefficient, field/covariance model, rank/null, and regularization fields	A weighting realization, any optimality label, and any variance meaning derived from the weight.

Family	Decision	Route unavailable before disposition
SCI-FLT-MATCHED-A0-002	Exact-operator numerical profile, separately authorized distinct scientific operator, or typed numerical-route unavailability	Any numerical- conformance statement.
SCI-FLT-MATCHED-A0-003	Complete, named-projected, or unavailable covariance scope, independently from resident or lineage representation	Covariance publication, standardization, or covariance-dependent use.
SCI-FLT-MATCHED-A0-004	Full, compact-exact, or lineage state representation preserving one query vocabulary	State audit/reanalysis and completion of products whose named use requires reconstructable state.
SCI-FLT-MATCHED-A0-005	Response domain/query/validity/consumer scope, then full, structured, or lineage representation	Response-dependent use and FLT-to-FRUIT producer-envelope use.
SCI-FLT-MATCHED-A0-006	Seven normative role verdicts and dependency graph, represented as separate, grouped, or vector records	Registered SCI-VAL named-use evaluation.

No option choice alone authorizes a numerical route. After owner disposition, an implementation still needs separate conformity evidence; profiles need separate SCI-VAL evaluation; observational validation, measured performance, readiness, and production remain separate decisions.

A Author-packet basis and firewall statement

Stage B r0.1 used only these eight admitted objects:

- AUTHOR_PACKET_MANIFEST.md and SCOPE_BRIEF.md;
- AUTHOR_SUPERSESSION_COVER.md and AUTHOR_CONVENTIONS_AND_OWNERSHIP.md;
- SCIENTIFIC_OWNER_DECISIONS_FOR_AUTHORSHIP.md;
- AUTHOR_OPERATOR_STATE_AND_PRODUCT_TAXONOMY.md; and
- AUTHOR_BOUNDARIES.md and REQUIRED_AUTHORED_OPTION_SETS.md.

The r0.2 revision additionally used only the scientific owner’s exact targeted closure directive dated 2026-08-31. Links in the packet were not opened. No implementation, configuration, schema, test, audit, validation, reduction, generated product, repository history, neighboring package, external source, or model-memory source supplied scientific content.